# Sara Hosseini

## Data Analyst · SQL · Python · Power BI · Statistical Analysis

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## PROFILE

Data Analyst with nearly seven years of KPI monitoring, performance analysis and data-quality work on large construction and infrastructure projects, and an M.Sc. in Artificial Intelligence from a German university (grade 1.8). I analyse data in SQL and Python, test results statistically, and present them in Power BI. My public portfolio includes an end-to-end product analytics case study (funnel, retention, cohorts, A/B testing) and probabilistic forecasting research. I bring the business side too: defining KPIs with stakeholders, finding the root cause when numbers disagree, and turning analysis into recommendations.

## TECHNICAL SKILLS

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|---|---|
| **Analytics** | SQL (CTEs, window functions, cohort and funnel queries) · Python (pandas, NumPy, SciPy) |
| **BI & Visualisation** | Power BI (data modelling, DAX measures, multi-page reports) · Matplotlib · Seaborn |
| **Statistics** | Hypothesis testing (z-test, Welch t-test, paired t-test) · confidence intervals · regression · forecast evaluation |
| **Applied Analytics** | KPI & performance analysis · conversion funnels · retention & cohort analysis · segmentation · A/B testing |
| **Predictive** | Time-series forecasting · quantile regression · scikit-learn · XGBoost · TensorFlow/Keras · PyTorch · NLP |
| **Data Quality & Tools** | Validation rules · root-cause analysis · pytest · Git · Jupyter · SQLite · Excel · SPSS |

## PROFESSIONAL EXPERIENCE

### Data & Reporting Analyst

Oct 2020 – Jul 2024

Baroj · Iran · construction and infrastructure projects

- Consolidated cost, schedule, procurement and progress data from multiple project sources into recurring KPI and management reports for several concurrent projects.
- Ran planned-vs-actual and trend analysis to show project managers where cost and schedule performance were drifting from plan.
- Investigated figures that disagreed across reporting sources, traced each discrepancy to its source data and applied validation checks before numbers reached management.
- Standardised KPI definitions, reporting templates and documentation across concurrent projects, keeping figures comparable between periods and teams.
- Gathered reporting requirements with project managers, engineers, finance and procurement and translated them into consistent management reporting.

### Junior Data & Reporting Analyst

Oct 2017 – Sep 2020

Abad Boom Pars Co. · Iran · construction projects

- Collected, validated and structured cost, schedule and progress data behind recurring project KPI reporting.
- Produced KPI summaries and progress reports for project managers; supported senior analysts with schedule and performance analysis.
- Resolved data discrepancies with engineering and project teams; maintained the reporting datasets and their documentation.

## ANALYTICS PROJECTS

**Product Analytics Case Study: Funnel, Retention & A/B Testing** 2026

Portfolio project · simulated European mobility app · github.com/Sara-Hosseini/product-analytics-case-study

- Generated and validated a synthetic dataset of 5,000 users and 141,813 behavioural events, covered by 23 validation checks and 25 automated pytest tests.
- Wrote 10 SQL analysis scripts (1,746 lines with schema) for KPIs, conversion funnel, D1/D7/D30 retention, monthly cohorts, segmentation and experiment evaluation.
- Found the main funnel leak: 52.1% of searching users never started a booking, while 86.5% of started bookings were completed.
- Evaluated an A/B test: conversion 26.89% → 30.45% (+3.55 pp, 95% CI 1.05–6.06 pp, p = 0.006); ARPU €7.02 → €8.80.
- Built a six-page Power BI report with 27 DAX measures (retention, funnel rates, ARPU and experiment uplift) and wrote prioritised business recommendations.

**Tools:** SQL (SQLite) · Python (pandas, NumPy, SciPy, Seaborn) · Power BI / DAX · pytest · Git

**Probabilistic Stock Price Forecasting (M.Sc. Thesis)** 2026

IU International University of Applied Sciences · github.com/Sara-Hosseini/financial-time-series-forecasting

- Collected, cleaned and aligned 11 years of daily prices for AAPL, MSFT and NVDA with about 1,200–1,350 news articles per stock from three sources.
- Built a Quantile LSTM producing 10th/50th/90th percentile forecasts, with news sentiment from FinBERT fine-tuned via LoRA (macro F1 0.31 → 0.80).
- Evaluated with five-fold walk-forward validation against XGBoost and naive baselines, using paired t-tests and Diebold-Mariano tests: 1-day intervals reached 78.8–83.0% coverage against an 80% target (XGBoost: 70.1%).

**Tools:** Python · pandas · TensorFlow/Keras · PyTorch · Hugging Face Transformers · XGBoost · scikit-learn

## EDUCATION

**M.Sc. Artificial Intelligence · Grade 1.8** Oct 2024 – Aug 2026

IU International University of Applied Sciences, Berlin · statistics, machine learning, deep learning, NLP, time-series forecasting, predictive analytics

**Master's degree, Business Administration (International Business)** Sep 2009 – Jul 2011

Alghadir Institute of Higher Education · recognised by ZAB as Master-level (Master im Fach Handelsmanagement) · survey research and analysis in SPSS

**B.Sc. Physics** Feb 2003 – Oct 2007

Urmia University

## CERTIFICATIONS

- Machine Learning Specialization, DeepLearning.AI & Stanford Online (Coursera)
- Pandas, Kaggle

## LANGUAGES

- English: C1 (professional)
- German: B1 (improving)
- Persian, Turkish: native